

(a)

$x = L_{bl}$

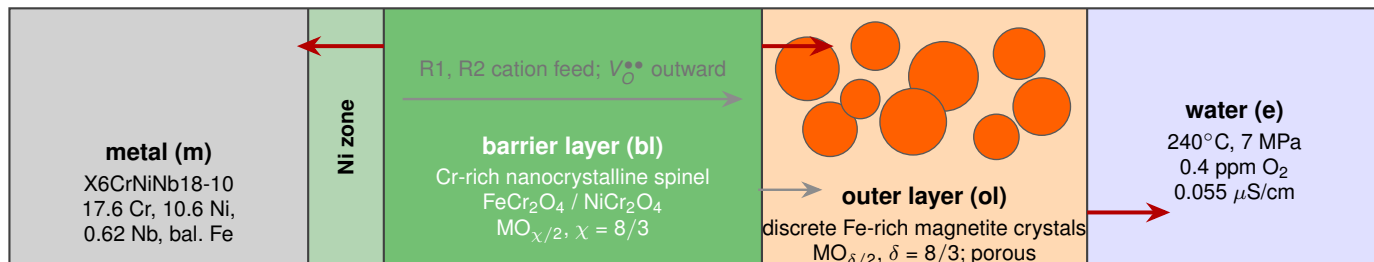
$x = 0$

$x = -L_{ol}$



R3 $m \rightarrow M_M + (\chi/2)V_O^{\bullet\bullet} + \chi e'$
barrier layer grows into the metal

R10, R10' **bl** $\rightarrow$ **ol** conversion
and proton-assisted bl dissolution



Ni: slow-oxidizing, accumulates by exclusion at m/bl; $dL_{Ni}/dt \approx 0$, not modeled

R4, R4', R5, R5' deposit outer-layer oxide at bl/ol

R11 **ol** dissolution / further oxidation to $MO_{\theta/2}$, $\theta = 3$

$\rightarrow$ moves a boundary (enters the thickness equations)

$\rightarrow$ lattice-conservative (no boundary displacement)

(b)